70 Days DSA Roadmap

Month 1: Pure Foundation Phase

Week 1: Arrays & Two-Pointer Techniques

Patterns
- Time & Space Complexity (deep)
- Arrays (prefix, suffix, difference array)
- Two pointer technique
- Sliding window basics

Practice Target
Total: **28–30 problems**
- Easy: 10
- Medium: 18–20
- Hard: 0–2 (optional)

Week 2: Strings + Hashing

Patterns
- HashMap / HashSet
- Frequency Count
- Anagram Patterns
- Substring Problems

Practice Target
Total: **30–35 problems**
- Easy: 8–10
- Medium: 20–22
- Hard: 3

Week 3 — Linked List + Stack + Queue

Patterns
- Fast & Slow Pointer
- Reverse Linked List
- Cycle Detection
- Monotonic Stack
- Stack / Queue Design

Practice Target
Total: **25–28 problems**
- Easy: 6–8
- Medium: 16–18
- Hard: 2–3

Week 4: Binary Search and Basic Recursion

Patterns
- Classic Binary Search Pattern
- First/Last Occurrence Pattern
- Search in Rotated Sorted Array
- Backtracking
- Divide and Conquer

Practice Target
Total: **24–28 problems**
- Easy: 6-8
- Medium: 14–16
- Hard: 1-2

Month 2: Core DSA Phase

Week 5: Trees and Hierarchical Traversal

Patterns
- Inorder traversal
- Preorder traversal
- Postorder traversal
- Level order traversal
- Height and diameter of tree
- Lowest common ancestor
- Path sum problems

Practice Target
Total: **24–28 problems**
- Medium: 20–22
- Hard: 4–6

Week 6: Ordered Structures and Priority Processing

Patterns
- Binary search tree operations
- Kth smallest element
- Kth largest element
- Priority queue fundamentals
- Top K frequent elements
- Median from data stream

Practice Target
Total: **20–23 problems**
- Medium: 16–18
- Hard: 4–5

Week 7: Graph Traversal

Patterns
- Breadth first search
- Depth first search
- Cycle detection in graph
- Topological ordering
- Disjoint set data structure
- Shortest path basics

Practice Target
Total: **26–29 problems**
- Medium: 20–22
- Hard: 6–7

Week 8: Greedy and Interval Problems

Patterns
- First/last occurrence
- Lower & upper bound
- Simple search on answers
- Subsets & permutations

Practice Target
Total: **16-18 problems**
- Medium: 14–15
- Hard: 2–3

Month 3: Advanced Problem Solving

Week 9: Dynamic Programming (1D)

Patterns
- Maximum subarray sum
- Longest increasing subsequence
- Coin change problems
- Knapsack style problems

Practice Target
Total: **22–25 problems**
- Medium: 18–20
- Hard: 4–5

Week 10: Advanced Dynamic Programming & Backtracking (2D)

Patterns
- Dynamic programming on strings
- Dynamic programming on grids
- Edit distance
- Word segmentation
- N-Queens problem

Practice Target
Total: **26–30 problems**
- Medium: 18–20
- Hard: 8–10

Week 11 Onward: Practice & Mock Interviews

Targets
- 8–10 full mock interviews
- 45-minute timed solving
- Verbal explanation practice

Revision Plan
- Re-solve 30 most important problems
- Revise all patterns once